ASSIGNMENT – 01

Entity–Relationship (ER) Diagram Design

Case Study: Online Learning Platform

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Assignment	Assignment – 01

Database Management System – Conceptual Design Assignment

1. Introduction / Problem Statement

The Online Learning Platform enables Instructors to create and publish Courses that are organised into Modules, which in turn contain Lessons, Assignments and Quizzes. Students browse Courses by Category, enrol after making a Payment, work through Lessons, submit Assignments, attempt Quizzes, write Reviews and earn Certificates on completion. An Admin oversees the platform. The system must therefore track user accounts and their roles, course content structure, enrolment and progress, assessment (assignments and quizzes with questions), payments, reviews and certification.

This document presents the conceptual data model of the system in the form of an Entity–Relationship (ER) diagram, lists every entity with its attributes and keys, defines the relationships with their cardinality and participation constraints, applies generalization/specialization to the user hierarchy, and finally maps the model to a relational schema suitable for implementation in a DBMS such as MySQL or PostgreSQL.

2. Entities and Attributes

Primary keys are underlined and shown in bold; foreign keys are italicised. All 17 entities identified from the case study are listed below.

Entity	Primary Key	Foreign Key(s)	Other Attributes
USER (superclass)	**UserID**	—	Name, Email, Password, Phone, DateJoined, RoleType
STUDENT (subclass)	**UserID (PK/FK)**	*UserID → User*	EnrollmentYear, EducationLevel
INSTRUCTOR (subclass)	**UserID (PK/FK)**	*UserID → User*	Qualification, Experience_Years, Bio
ADMIN (subclass)	**UserID (PK/FK)**	*UserID → User*	AccessLevel
CATEGORY	**CategoryID**	—	CategoryName, Description
COURSE	**CourseID**	*CategoryID → Category, InstructorID → Instructor*	Title, Description, Price, Level, Duration, CreatedDate
MODULE	**ModuleID**	*CourseID → Course*	Title, SequenceNo
LESSON	**LessonID**	*ModuleID → Module*	Title, VideoURL, Duration
ASSIGNMENT	**AssignmentID**	*ModuleID → Module*	Title, DueDate, MaxMarks
QUIZ	**QuizID**	*ModuleID → Module*	Title, TotalMarks
QUESTION	**QuestionID**	*QuizID → Quiz*	QuestionText, Options, CorrectAnswer
ENROLLMENT	**EnrollmentID**	*StudentID → Student, CourseID → Course*	EnrollmentDate, Status, ProgressPercent
SUBMISSION	**SubmissionID**	*StudentID → Student, AssignmentID → Assignment*	SubmissionDate, FileURL, MarksAwarded
QUIZ_ATTEMPT	**AttemptID**	*StudentID → Student, QuizID → Quiz*	Score, AttemptDate
PAYMENT	**PaymentID**	*StudentID → Student, CourseID → Course*	Amount, PaymentDate, PaymentMethod, Status

Entity	Primary Key	Foreign Key(s)	Other Attributes
REVIEW	**ReviewID**	*StudentID → Student, CourseID → Course*	Rating, Comment, ReviewDate
CERTIFICATE	**CertificateID**	*StudentID → Student, CourseID → Course*	IssueDate, CertificateURL

3. Relationships, Cardinality and Participation Constraints

Double-line participation in the diagram indicates total (mandatory) participation; single-line indicates partial (optional) participation, matching the table below.

Relationship	Entities Involved	Cardinality	Participation	Description
Teaches	Instructor – Course	**1 : N**	Instructor: Partial Course: Total	An instructor may teach many courses; every course must have exactly one instructor.
BelongsTo	Category – Course	**1 : N**	Category: Partial Course: Total	A category groups many courses; every course must belong to one category.
HasModule	Course – Module	**1 : N**	Course: Total Module: Total	Every course is structured into one or more modules; every module belongs to exactly one course.
HasLesson	Module – Lesson	**1 : N**	Module: Total Lesson: Total	Every module has one or more lessons; every lesson belongs to one module.
HasAssignment	Module – Assignment	**1 : N**	Module: Partial Assignment: Total	A module may have zero or more assignments; every assignment belongs to one module.
HasQuiz	Module – Quiz	**1 : N**	Module: Partial Quiz: Total	A module may have zero or more quizzes; every quiz belongs to one module.
Contains	Quiz – Question	**1 : N**	Quiz: Total Question: Total	Every quiz contains one or more questions; every question belongs to one quiz.
EnrollsIn	Student – Course (via Enrollment)	**M : N**	Student: Partial Course: Total	A student enrols in many courses and a course has many students; resolved by the associative entity Enrollment.
Submits	Student – Assignment (via Submission)	**M : N**	Student: Partial Assignment: Partial	A student submits many assignments; an assignment receives many submissions.
Attempts	Student – Quiz (via Quiz_Attempt)	**M : N**	Student: Partial Quiz: Partial	A student attempts many quizzes; a quiz is attempted by many students.
Makes / For	Student – Payment – Course	**1:N / N:1**	Student: Partial Payment: Total	A student makes many payments, each payment is for exactly one course.
Writes	Student – Course (via Review)	**M : N**	Student: Partial Course: Partial	A student writes reviews for the courses taken; a course receives reviews from many students.

5. Complete ER Diagram

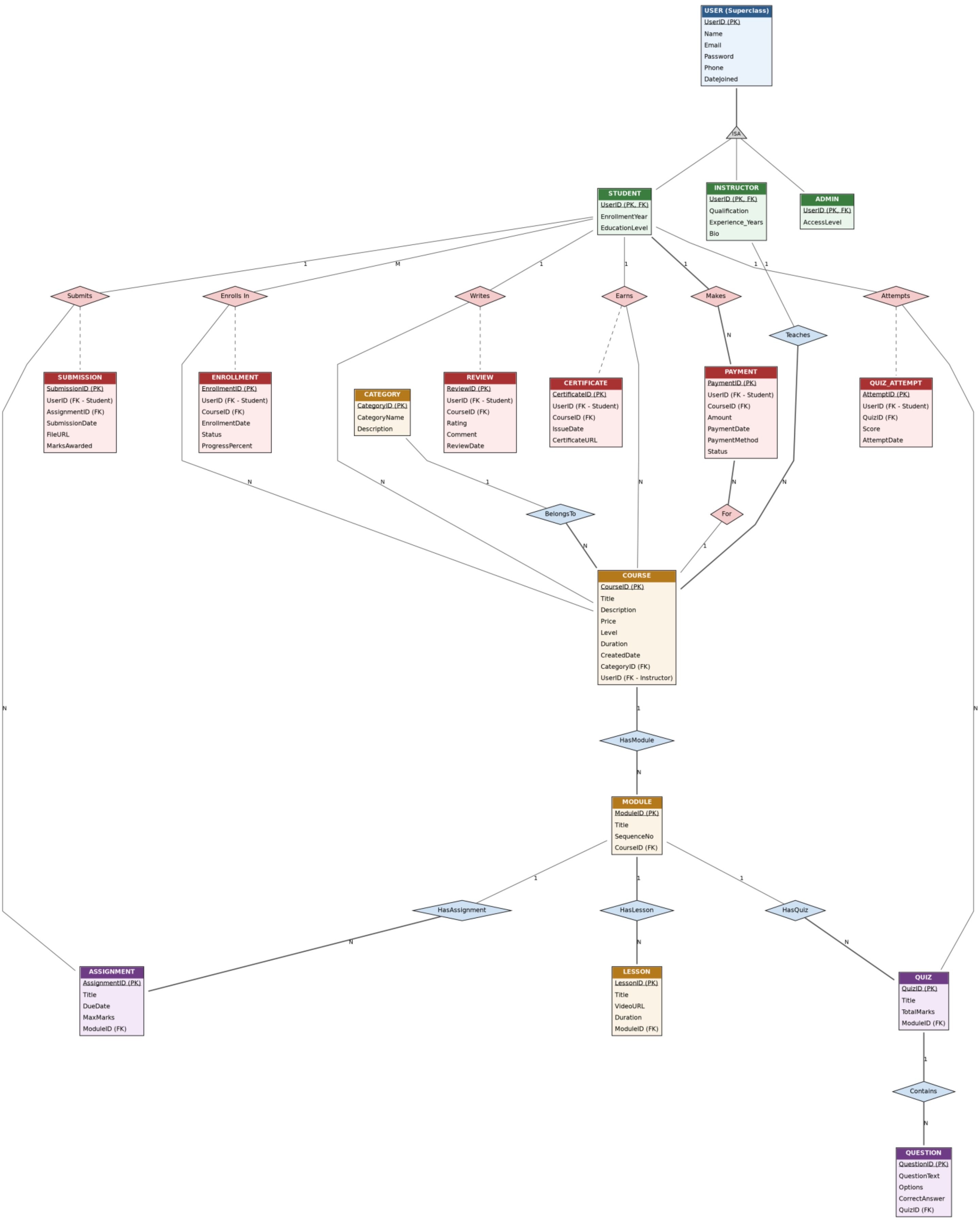

Figure 1: ER diagram of the Online Learning Platform (Chen-style, table notation). Underlined attribute = Primary Key; diamond = relationship; triangle = ISA generalization; double line = total participation.

6. Relational Schema (Mapping to Tables)

Applying standard ER-to-relational mapping rules (each strong entity → table with its PK; each weak/associative entity → table with a composite/FK-based PK; 1:N relationships → FK placed on the 'many' side; M:N relationships already resolved via associative entities; generalization → one table per subclass sharing the superclass PK) gives the following schema:

User (UserID PK, Name, Email, Password, Phone, DateJoined, RoleType)

Student (UserID PK/FK→User, EnrollmentYear, EducationLevel)

Instructor (UserID PK/FK→User, Qualification, Experience_Years, Bio)

Admin (UserID PK/FK→User, AccessLevel)

Category (CategoryID PK, CategoryName, Description)

Course (CourseID PK, Title, Description, Price, Level, Duration, CreatedDate, CategoryID FK→Category, InstructorID FK→Instructor)

Module (ModuleID PK, Title, SequenceNo, CourseID FK→Course)

Lesson (LessonID PK, Title, VideoURL, Duration, ModuleID FK→Module)

Assignment (AssignmentID PK, Title, DueDate, MaxMarks, ModuleID FK→Module)

Quiz (QuizID PK, Title, TotalMarks, ModuleID FK→Module)

Question (QuestionID PK, QuestionText, Options, CorrectAnswer, QuizID FK→Quiz)

Enrollment (EnrollmentID PK, StudentID FK→Student, CourseID FK→Course, EnrollmentDate, Status, ProgressPercent) — UNIQUE(StudentID, CourseID)

Submission (SubmissionID PK, StudentID FK→Student, AssignmentID FK→Assignment, SubmissionDate, FileURL, MarksAwarded)

Quiz_Attempt (AttemptID PK, StudentID FK→Student, QuizID FK→Quiz, Score, AttemptDate)

Payment (PaymentID PK, StudentID FK→Student, CourseID FK→Course, Amount, PaymentDate, PaymentMethod, Status)

Review (ReviewID PK, StudentID FK→Student, CourseID FK→Course, Rating, Comment, ReviewDate) — UNIQUE(StudentID, CourseID)

Certificate (CertificateID PK, StudentID FK→Student, CourseID FK→Course, IssueDate, CertificateURL)

7. Conclusion

The ER model captures the complete structural and functional scope of the Online Learning Platform: the User generalization hierarchy, the content hierarchy (Course → Module → Lesson/Assignment/Quiz → Question), the transactional entities (Enrollment, Submission, Quiz_Attempt, Payment, Review, Certificate), and the cardinality/participation constraints governing each relationship. The model is fully normalised (up to 3NF/BCNF, since every non-key attribute depends only on its entity's primary key) and maps directly to the 17-table relational schema in Section 6, which can be implemented directly in any relational DBMS such as MySQL, PostgreSQL or Oracle.